

Key open source packages in Python for machine learning

There are many open source packages in Python for addressing ML problems, and can be used based on the nature of the issue.

NumPy: This library provides support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays.

It handles linear algebra, Fourier transforms, and random numbers. It interoperates with other libraries like Matplotlib, SciPy, scikit-learn and TensorFlow. NumPy is widely used for handling sound waves, images, and other binary functions

Scikit-learn: This provides a broad range of machine learning functions like classification, regression, clustering, dimensionality reduction, model selection, and preprocessing.

The scikit-learn developer team maintains a strong focus on code quality and comprehensive documentation with the help of the open source community. It can interoperate with the numeric and scientific libraries of Python like NumPy and SciPy.

Theano: This scientific computing library allows defining, optimising, as well as evaluating mathematical expressions. Theano helps make data-intensive calculations up to a hundred times faster than when executing these on the CPU alone.

Well-optimised for GPUs, it offers effective symbolic differentiation and includes extensive code-testing capabilities. Theano can automatically avoid errors and bugs when dealing with logarithmic and exponential functions, and has built-in tools for unit testing and validation.

TensorFlow: An end-to-end library for performing high-end numerical computations, it can handle deep neural networks for image recognition, handwritten digit classification, recurrent neural networks, NLP (natural language processing), word embeddings and PDE (partial differential equation).

It supports a variety of different toolkits for constructing models at varying levels of abstraction. Its flexible architecture helps it run on CPUs, GPUs, and TPUs.

Keras: Used for constructing neural networks and machine learning projects, Keras interoperates with TensorFlow, Theano and R, and can run efficiently on CPUs and GPUs.

It offers standalone modules including optimisers, neural layers, activation functions, initialisation schemes, cost functions, and regularisation schemes. Keras functions as a

user-friendly, extensible interface that enhances modularity and total expressiveness.

PyTorch: This supports computer vision, machine learning, and natural language processing. It integrates well with the Python data science stack, including NumPy.

It has a robust framework to build computational graphs on the go and even change them in runtime. It supports Tensor computations with GPUs, and helps with performance optimisation and scalable distributed training in research as well as production.

Pandas: It offers a wide range of tools for data manipulation and analysis. Using this highly stable library, we can read data from various data sources like CSV, SQL databases, JSON files, and Excel. It can handle tabular data, time series data, matrix data, etc, and has two types of data structures — data frames and series.

XGBoost: This optimised gradient-boosting library for tabular data delivers strong performance and scalability in classification and regression tasks.

MLflow: This tool for the ML lifecycle is used for experiment tracking, model registry, and deployment workflows.

ONNX: This is another tool for the ML lifecycle, which is used for model interchange and runtime portability across frameworks.

Industry use cases of Python-based AI/ML

Healthcare: Python-based ML applications help to monitor and predict diseases, detect injuries, scan health parameters of patients regularly, etc. ML is used to discover new drugs and help diagnose diseases more accurately, apart from the automation of administrative tasks and patient management systems.

Banking and finance: Fintech industries stand to gain much by using AI and ML applications because these can help in customising the user experience and detecting fraud, two major concerns of this industry.

In addition, ML is used to detect customer attrition, loan defaulter prediction, fraud detection, credit scoring, money laundering prevention, and portfolio management.

E-commerce: AI and ML-based solutions help businesses to create a better customer experience and increase their sales revenue. ML helps in content personalisation, chatbots for improving performance, dynamic pricing, identifying shoppers' data patterns, and predicting how